

Shanav Bagga

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EDUCATION

University of Illinois at Urbana-Champaign

Champaign, IL

Master's in Computer Science

BS May 2025, MCS May 2026

Relevant Courses: Data Structures, Distributed Systems, Databases, Data Mining, Natural Language Processing, Algorithms, Computer Architecture, Computer Networks, Cloud Computing Applications, Cloud Storage Systems

EXPERIENCE

College Board

July 2026 - Present

Software Engineer I

New York, NY

- Developed an end-to-end test harness to validate cell and tenant provisioning workflows for the control plane of a cell-based architecture, executing locally with LocalStack and in AWS

Walgreens

June 2025 - August 2025

Software Engineer Intern

Deerfield, Illinois

- Implemented a full-stack web app to optimize the creation and distribution of change request deployment plans through Node.js, React.js, and Express.js, eliminating 1000+ email chains
- Created Excel workbooks for each deployment plan from user form input, stored the files in Azure Blob Storage, stored metadata in Azure SQL Server, and hosted the app in Azure App Services
- Spearheaded the migration of Dynatrace OneAgent from 250 host servers and 6 tenants to a consolidated prod and non-prod tenant environment, utilizing multiprocessing to execute in parallel

Arora Innovations

June 2024 – August 2024

Software Engineer Intern

Remote

- Fine-tuned a Stable Diffusion and Control Net model on 1000 image pairs for 5 epochs, optimizing hyperparameters to achieve a 60% improvement in output image quality for avatar creation
- Implemented a robust REST API through Node.js and Express.js and deployed it to Google Compute Engine's GPU infrastructure, achieving a 40% reduction in latency within three weeks

Siebel School of Computing & Data Science

August 2023 – December 2023

Undergraduate Teaching Assistant

Champaign, IL

- Held weekly office hours assisting 50+ students understanding key C++ concepts such as compilation, stacks, memory, queues, trees, graph theory, smart pointers, polymorphism, and file streams
- Led 14 lab sessions which enhanced student comprehension and project readiness through coding exercises

AI Camp

May 2023 – August 2023

Software Engineer Intern

Remote

- Developed a Python chatbot hosted with Streamlit which allows users to select from 3 different query engines and upload up to 10 files simultaneously, enabling comprehensive search capabilities
- Engineered an automated testing suite of 1,000 queries using OpenAI's API and reduced the hallucination rate to 5% and a median accuracy rate of 90%, significantly enhancing the chatbot's reliability
- Strengthened performance and reached a 90% median response accuracy score by integrating the Langchain and Pinecone pipeline, which successfully decreased the total incorrect responses by 25%

PROJECTS

RainStorm | Go

December 2025

- Built a distributed stream processing system supporting filter, transform, and aggregate operators across 10 worker VMs with a leader-based scheduler and hash-partitioned tuple routing
- Configured auto-scaling via a Resource Manager that monitors per-task tuple throughput and dynamically scales tasks up/down against high/low watermarks
- Benchmarked against Apache Spark across 4 workloads, achieving comparable end-to-end latency on synthetic datasets

Customer ETL Pipeline | Python, MySQL, Spark, Google BigQuery

May 2024

- Implemented an ETL Pipeline that extracted 250,000 records of ecommerce customer data from MySQL, transformed the data in PySpark through filtering and aggregating by customer, and loaded it to Google BigQuery
- Developed a logistic regression model to predict customer retainment, achieving an accuracy rate of 85%

TECHNICAL SKILLS

Languages: C/C++, Python, Java, Go, Javascript, Typescript, Rust, SQL, R, HTML, CSS

Developer Tools: Git, Docker, MySQL, AWS, Linux, GCP, Azure, MongoDB, Kubernetes, Terraform

Frameworks: React.js, Flask, Spark, Node.js, Express.js, Tensorflow, PyTorch